

# DINISH S K

## ROBOTICS SOFTWARE DEVELOPER

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### SUMMARY

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Robotics Software Developer with strong proficiency in Navigation2 and Moveit2 stack in ROS2. Expert level in C++ and Python. Successfully developed and contributed to multiple mobile robotics projects and strong problem-solving abilities. Currently seeking an opportunity to apply my skills in a professional setting and contribute to robotics solutions.

### EXPERIENCE

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#### Associate Robotics Engineer

Goat Robotics Private Limited, Coimbatore

Feb 2026 - Present

#### Freelance Developer

Self Employed

Dec 2024 - Jan 2026

- Worked with clients across the world to develop robotics solutions and ROS2 applications.
- Identify the problems faced by client while development of the real-world robot and provide them accurate solutions.

### EDUCATION

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#### B.E. in Robotics and Automation

United Institute of Technology, Coimbatore

Aug 2020 - May 2024

GPA: 8.88

#### HSC in Tamil Nadu State Board

Universal Matriculation Higher Secondary School, Tirupur

Jun 2019 - Apr 2020

GPA: 9.27

#### SSLC in Tamil Nadu State Board

Universal Matriculation Higher Secondary School, Tirupur

Jun 2017 - Apr 2018

GPA: 9.40

### PROJECTS

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#### Autonomous Geo - Exploration Rover

Jan 2024 – May 2024

- Tele-operated rover powered by Arduino Mega for efficient nutrient analysis of the agriculture land.
- Collects the statistical data and provide the suggestions for the efficient farming and irrigation.
- Low cost – Long Range rover helps to monitor and analysis the health of crops.
- Being responsible for the software stack, integrated the LoRa, GPS and NPK Sensor, and developed the real-time monitoring dashboard.
- Apart from software, worked with designing and electronics.

#### Autonomous Mobile Vacuum Cleaning Robot

Aug 2023 – Dec 2023

- A Mobile Robot with LiDAR and ROS2 for Vacuum Cleaning in Indoor environments.
- Low cost robot for efficient mapping and navigation even in the messy environment.
- Being responsible for the software stack, utilized the Nav2 Stack in ROS2, with RaspberryPi for autonomous navigation.
- Apart from RaspberryPi, used Arduino UNO for integrating the Encoders and Drive motors with Synchronous communication between Arduino and RaspberryPi.
- Also worked in architecting and designing the robot.

#### Colour Based Product Separator

Dec 2022

- A conveyor end fitting for mechanical separation of the products based on colour.
- Integration of the servo and colour sensor with Arduino UNO enables detection of colour and synchronize the separation.

## TECHNICAL SKILLS

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### Programming

[Python](#)  
[C++](#)  
[Java](#)

### Robotics Frameworks

[ROS2](#)  
[Navigation2](#)  
[Moveit2](#)

### Design & Simulation

[Fusion360](#)  
[SolidWorks](#)  
[Gazebo & Issac Sim](#)

### Algorithms

[Data Structures](#)  
[Navigation Algorithms](#)  
[SLAM](#)

### Development Tools

[Git](#)  
[Docker](#)

## NON-TECHNICAL SKILLS

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[Problem Solving](#)  
[Analytical Thinking](#)  
[Quick Learner](#)

## COURSES & CERTIFICATIONS

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### Experimental Robotics

[NPTEL Online Certifications](#)  
Secured Silver + Elite Certificate with 87%

[Mar 2024](#)

### ROS2 and Mobile Robots

[Newton's Innovation Lab](#)

[Nov 2023](#)

### Foundations of Cognitive Robotics

[NPTEL Online Certifications](#)  
Secured Elite Certificate with 70%

[Aug 2023](#)

### Wheeled Mobile Robots

[NPTEL Online Certifications](#)  
Secured Gold + Elite Certificate with 90%, ranked in the top 5% of the course.

[Mar 2023](#)

### Mechanics and Control of Robotic Manipulators

[NPTEL Online Certifications](#)  
Secured Silver + Elite Certificate with 82%, ranked as the course topper.

[Sep 2022](#)

### Python

[GUVI](#)

[Feb 2022](#)

## LANGUAGES

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English

[Proficient](#)

Tamil

[Native](#)